

WALEED IQBAL

Senior Software Engineer | Agentic AI Evaluation and Agent Infrastructure

Faisalabad, Pakistan | Open to relocation with visa sponsorship | Available remote across US and EU hours

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PROFILE

Senior software engineer specialising in agentic AI evaluation infrastructure. I build the systems that measure whether AI agents actually work: provider agnostic execution across frontier and self hosted models, reproducible sandboxed environments on GKE and AWS, layered automated grading, and the distributed orchestration that keeps long running agent runs healthy. Five years spanning production SaaS delivery and AI infrastructure, promoted from associate to senior engineer before moving into evaluation platform work, delivered remotely across time zones with colleagues and clients on several continents.

CORE EXPERTISE

Agent systems and evaluation: computer use agents, multi model evaluation, LLM as judge, rubric design, trajectory analysis, automated review, Playwright, OSWorld, WebArena, BrowseComp, Harbor

Backend and orchestration: Python, FastAPI, Django, Celery, Redis, PostgreSQL, REST APIs, asynchronous and distributed systems, Ruby on Rails, Node.js

Cloud and infrastructure: GCP, GKE, AWS, EC2, Kubernetes, Docker, prebaked golden volumes, GCS, Prometheus, Flower, GitLab CI/CD

Frontend and tooling: React, Next.js, TypeScript, Electron, Chrome extensions, FFmpeg, WebRTC, vLLM

PROFESSIONAL EXPERIENCE

Turing — Python Developer, Agentic AI Evaluation Platform

Sep 2023 - Present

Remote, full time contract | San Francisco headquartered company, working within a globally distributed team

- Collapsed five provider integrations into one execution loop, so any frontier or self hosted model runs the same task without code changes and new providers land without regressing existing ones, by normalising divergent tool calling formats, action vocabularies and screenshot handling into a single trajectory format (OpenAI, Anthropic, Gemini, OpenRouter, vLLM).
- Cut the false verdicts that single signal grading produces, where LLM judges drift between runs, DOM checks break on cosmetic markup and exact matching punishes agents that solve a task by a different valid route, by combining golden trajectory comparison, LLM evaluators, DOM and state assertions and graders running inside the target environment behind 18+ assertion operators.
- Sustained 30+ concurrent long running agent tasks with zero duplicate dispatch and automatic recovery from worker loss, via database governed dispatch using atomic claims, PostgreSQL advisory locks, Redis leases and watchdog sweeps across a 17 queue, 14 worker Celery fleet.
- Removed environment drift as a source of false results by booting every desktop run from prebaked golden volumes, cutting startup time and giving engineers VPN access to debug live agent runs instead of inferring failures from logs.
- Made iterating on grading logic close to free by snapshotting machine state, so a changed verifier is retested against saved state instead of paying for a fresh model run and a new VM on every attempt.
- Put browser and full desktop agent runs on a single execution path across GKE and AWS, and made every run reviewable afterwards by capturing session video, screenshots, action timelines and state traces centrally, covering OSWorld, WebArena, BrowseComp and Harbor workloads.
- Made every client delivery traceable to an exact signed off version, replacing ad hoc execution with a role gated lifecycle built on append only versioning and full audit events.
- Brought an evaluation platform from nothing to production in three months, now the daily tool for several hundred people across task authoring, model runs, trajectory review and client delivery, with a task library grown into the thousands.
- Established what each task actually requires rather than what the author assumed, by recording ideal human runs as video and raw event streams and turning them into the ground truth that prompts, rubrics and automated graders are written against.
- Took engineering out of the delivery loop, letting operations and research export multi gigabyte result sets themselves through signed URLs and streaming ZIP, and made per run model spend and pass rates visible without asking an engineer.
- Shipped new evaluation capabilities without destabilising production by proving each one as a standalone MVP before folding it into the main harness.

Softaims — Senior Software Engineer

Aug 2021 - Jul 2024

Lahore, Pakistan | Promoted twice: Associate (2021) to Software Engineer (2022) to Senior (2024)

Final 10 months held concurrently with the Turing contract, with both employers' knowledge.

- Directed architecture, delivery planning and release decisions across three concurrent client products with a team of three engineers, reporting production risk directly to the clients themselves.
- Cleared Shopify App Store review with an upsell app rebuilt from a five year old codebase, adding theme app extensions, recommendations driven by order history and stated customer demand, and a GraphQL rewrite that kept the app current through Shopify's API transition.
- Dropped latency on a crypto platform buckling under server load by moving realtime third party data fetching onto Redis backed workers with caching, so the main backend service stopped absorbing traffic it could not carry.
- Handed release control to a non technical client managing a large network of German institutes: every merged branch deployed itself to a review environment he checked and signed off on, then reached production on a button click, by rebuilding delivery on GitLab CI/CD with Dockerised builds shipping to Heroku after retiring the previous GitHub based workflow.
- Let every institute on that platform run under its own subdomain on a shared Rails and React codebase, and rebuilt the interface on Tailwind while integrating the APIs the institutes are administered through.
- Gave support agents live sight of a customer's screen through an embeddable WebRTC tool with video and audio calling, so problems were diagnosed in session instead of over email, and shipped a companion Chrome extension recording screen captures straight to storage in three formats via FFmpeg.

SELECTED PROJECTS

- AI chatbot platform for Shopify merchants (Node.js, TypeScript, Shopify Remix, Prisma, PostgreSQL with pgvector, BullMQ, OpenRouter): merchants connect a store over OAuth and complete a one time configuration quiz that generates a per store system prompt. The assistant answers shopper questions through real time tool calls against store data rather than bulk context injection, with retrieval over pgvector, order lookup gated by order number and email verification for unauthenticated shoppers, and audience scoping so shoppers never receive merchant operational data. Instrumented with per session cost telemetry covering cost per call, call histories and reasoning versus tool calling breakdown. Working build, store listing in preparation.
- Personalized clothing recommendation engine (Python, machine learning): click based ranking that blends user preferences with all time, weekly, daily and hourly interaction windows to estimate product affinity.
- Veritas, social reading platform, final year project (Android, React, Node.js, Firestore): Kindle style in app reading combined with Goodreads style discovery, reader profiles and an interest and engagement based recommendation approach.
- Technical writing on agent evaluation: published an architecture and methodology overview covering provider abstraction, layered verification and reproducible sandboxed environments at github.com/iamwaleediqbal.

EDUCATION

BSc Computer Science, FAST National University of Computer and Emerging Sciences, Lahore 2017 - 2021

LANGUAGES

English (professional working proficiency) | Urdu (native) | Punjabi (native) | Hindi (spoken and understood, not written)